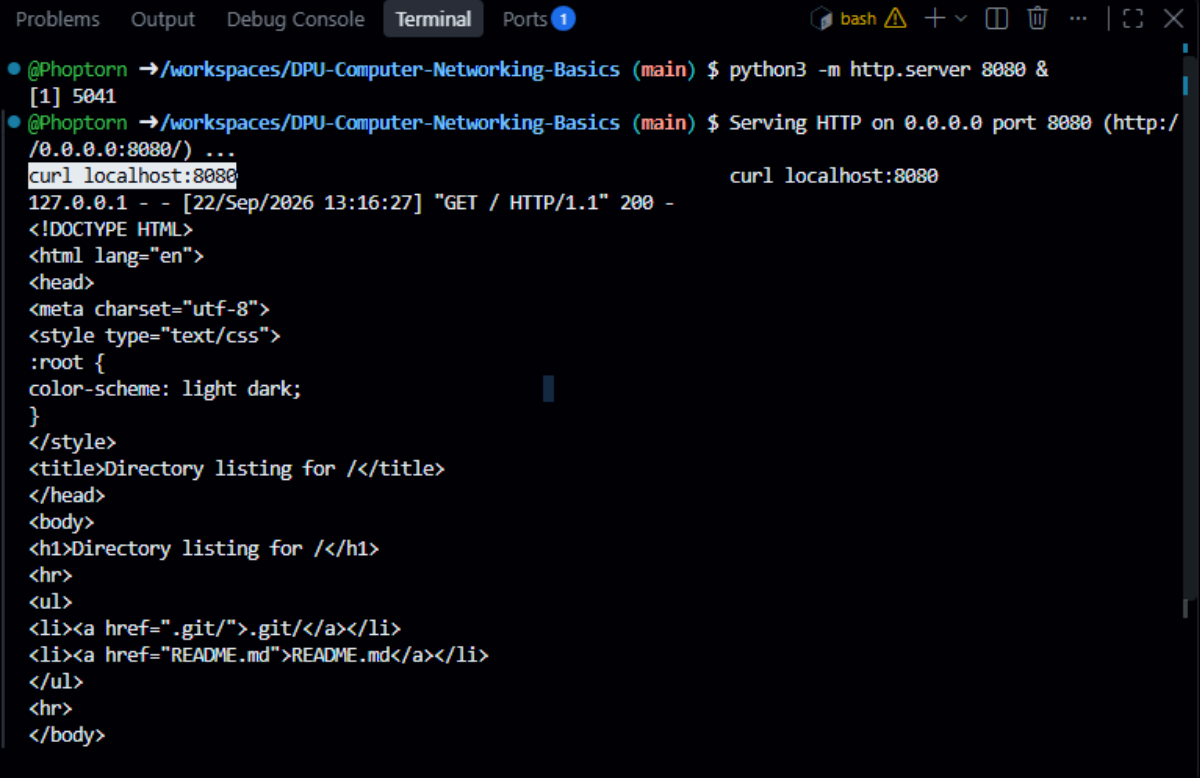


## Module 2 — Computer Networking Basics: Exercises

### Exercise 1 — Be the client AND the server



```
Problems Output Debug Console Terminal Ports 1
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $ python3 -m http.server 8080 &
[1] 5041
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $ Serving HTTP on 0.0.0.0 port 8080 (http://
0.0.0.0:8080/) ...
curl localhost:8080 curl localhost:8080
127.0.0.1 - - [22/Sep/2026 13:16:27] "GET / HTTP/1.1" 200 -
<!DOCTYPE HTML>
<html lang="en">
<head>
<meta charset="utf-8">
<style type="text/css">
:root {
color-scheme: light dark;
}
</style>
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href=".git/">.git</a></li>
<li><a href="README.md">README.md</a></li>
</ul>
<hr>
</body>
```

Question: In the first command you started a server. In the second, which role were you playing?  
> Client to view the website.

### Exercise 2 — Subnet mask bit calculation

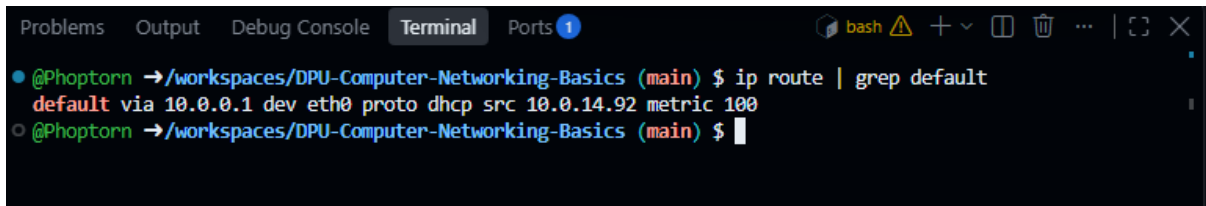
/24 ->  $2^8 - 2 = 254$

/16 ->  $2^{16} - 2 = 65,534$

/28 ->  $2^4 - 2 = 14$

/26 ->  $2^6 - 2 = 62$

## Exercise 3 — Find your default gateway



```
Problems Output Debug Console Terminal Ports 1
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $ ip route | grep default
default via 10.0.0.1 dev eth0 proto dhcp src 10.0.14.92 metric 100
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $
```

Question: What IP address is listed as your default gateway? Is it on the same subnet as your own IP (from ip a)?

> Default gateway: 10.0.0.1. Yes, they are on the same subnet.

## Exercise 4 — Run two servers, one machine



```
Problems Output Debug Console Terminal Ports 2
@Phoptorn →/workspaces/DPL
80 &
python3 -m http.server 8081
ss -tulnp | grep python
[1] 22092
[2] 22103
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $ Serving HTTP on 0.0.0.0 p
ort 8080 (http://0.0.0.0:8080/) ...
Serving HTTP on 0.0.0.0 port 8081 (http://0.0.0.0:8081/) ...
█
```

Question: How many ports show up listening? Try curl localhost:8080 and curl localhost:8081 — two totally separate conversations, same machine.

Clean up after with kill %1 and kill %2.

> have 2 port listening, if we kill the process server will not be unable to connect to the server.

## Exercise 5 — Build & break down a real URL

```
Problems Output Debug Console Terminal Ports 1 bash - demo + - [ ] [ ] ... | [ ] [ ] X
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $ mkdir demo && cd demo
echo "hello" > predict.txt
python3 -m http.server 8080 &
curl localhost:8080/predict.txt
[1] 27158
curl: (7) Failed to connect to localhost port 8080 after 0 ms: Couldn't connect to server |
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics/demo (main) $ Serving HTTP on 0.0.
0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [22/Sep/2026 14:04:34] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [22/Sep/2026 14:04:36] "GET /predict.txt HTTP/1.1" 200 -
|
```

### Directory listing for /

- 
- [predict.txt](#)
- 

Question: What happens if you run curl  
localhost:8080/missing.txt instead?

> If we run curl localhost:8080/missing.txt the answer is below.

### Error response

Error code: 404

Message: File not found.

Error code explanation: 404 - Nothing matches the given URI.

## Exercise 6 — Watch DNS resolve

```
Problems Output Debug Console Terminal Ports bash + - [ ] [ ] ... [ ] [ ] X
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $ dig example.com
bash: dig: command not found
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $
```

Question: Find the line with a number followed by IN A and an IP address — that number is the TTL in seconds. How long will resolvers cache this answer before asking again?

> bash: dig and ping command not found, because I didn't install a package.

## Exercise 7 — Check your own network config

```
Problems Output Debug Console Terminal Ports bash + - [ ] [ ] ... [ ] [ ] X
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 70:a8:a5:72:b9:16 brd ff:ff:ff:ff:ff:ff
    inet 10.0.14.92/16 metric 100 brd 10.0.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::72a8:a5ff:fe72:b916/64 scope link
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:5f:79:3a:b3 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $
```

Question: Find your Codespace's own IP address and subnet mask in the output. Does ip a show your machine's IP, or your gateway's IP?

> 10.0.14.92/16

/16 = Subnet mask 255.255.0.0

## Exercise 8 — Run the full network toolkit

```

Problems Output Debug Console Terminal Ports
@Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) $ ping -c 4 example.com
curl -I https://example.com
ss -tulnp
bash: ping: command not found
HTTP/2 200
date: Tue, 22 Sep 2026 14:36:37 GMT
content-type: text/html
server: cloudflare
last-modified: Tue, 15 Sep 2026 23:41:26 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 1
cf-cache-status: HIT
cf-ray: a3f20b009955a8c1-SIN

Netid  State      Recv-Q      Send-Q      Local Address:Port      Peer Address:Port
Process
udp    UNCONN     0            0            127.0.0.1:53             0.0.0.0:*
udp    UNCONN     0            0            10.0.14.92:68            0.0.0.0:*
tcp    LISTEN     0            128          0.0.0.0:2222             0.0.0.0:*
tcp    LISTEN     0            4096         127.0.0.1:44375          0.0.0.0:*

```

- @Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) \$ curl https://example.com  
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>
- @Phoptorn →/workspaces/DPU-Computer-Networking-Basics (main) \$ curl -I https://example.com  
HTTP/2 200  
date: Tue, 22 Sep 2026 14:38:32 GMT  
content-type: text/html  
server: cloudflare  
last-modified: Tue, 15 Sep 2026 23:41:26 GMT  
allow: GET, HEAD  
accept-ranges: bytes  
age: 12842  
cf-cache-status: HIT  
cf-ray: a3f20dd01844fcea-SIN

Question: What does curl -I show, compared to a plain curl?

(Hint: -l fetches headers only, not the whole page.)

> shows only the HTTP response headers (status code, content type, server info without the content).